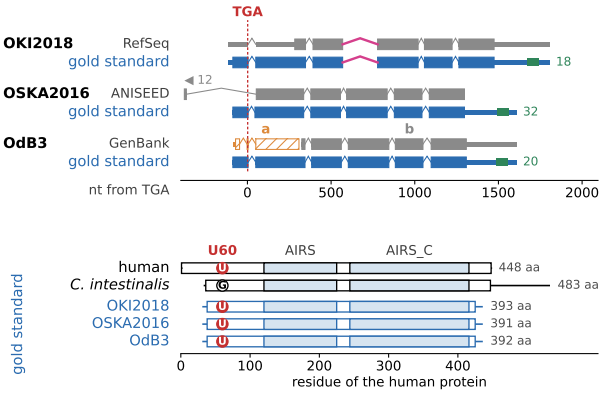
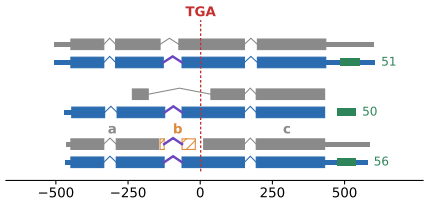
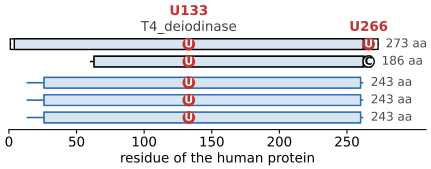
SEPHS2



OKI2018 RefSeq XP_081635792.1: coding sequence starts 79 codons after the TGA.
OSKA2016 ANISEED S25.g14713.01.t: TGA exon skipped; fused with the upstream gene.
OdB3 GenBank a CBY08599.1: other frame; b CBY08598.1: starts at an internal ATG.
proteins human Q99611; C. intestinalis Cint_SEPHS2.5.

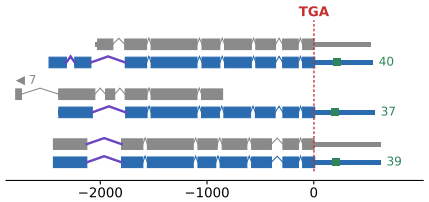
DIO_PAR

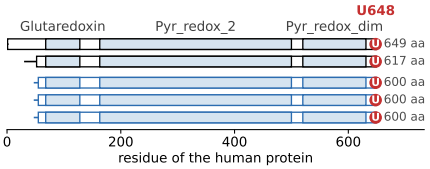




OKI2018 RefSeq XP_081636384.1: selenocysteine declared; one intron 11 nt away from the GA-AG intron of the transcripts.
OSKA2016 ANISEED S2.g03083.01.t: TGA in an intron of the annotated coding sequence.
OdB3 GenBank a CBY14356.1: ends before the TGA; b CBY14355.1: other frame; c CBY14354.1: starts after the TGA. Gold standard: protein from the ESTs (one extra base in the assembly).
proteins human Q92813; C. intestinalis Cint_DIO.2.

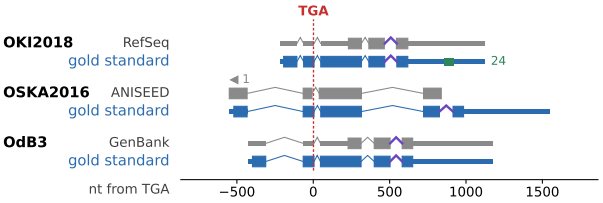
TXNRD

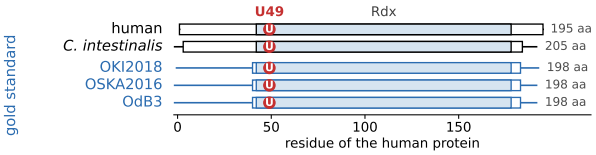




OKI2018 RefSeq XP_081643508.1: N-terminus from other exons (no glutaredoxin domain); TGA read as stop.
OSKA2016 ANISEED S8.g08759.01.t: extra exon in the GA-AG intron; ends in an intron before the TGA.
OdB3 GenBank CBY19735.1: one intron differs from the ESTs; TGA read as stop. Gold standard: exons beyond the ESTs from the GenBank model.
proteins human Q16881; C. intestinalis Cint_TXNRD.14.

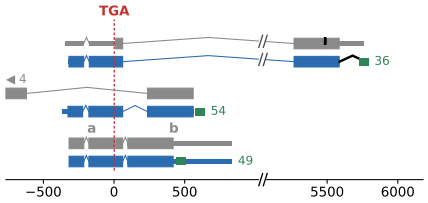
SELENOT

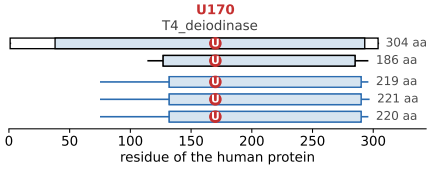




OKI2018 RefSeq XP_081643256.1: same exon chain; coding sequence starts at an internal ATG, 62 codons after the TGA.
OSKA2016 ANISEED S10.g09589.01.t: TGA in an intron of the coding sequence; extra 5′ exon; ends in the GA-AG intron.
OdB3 GenBank CBY23255.1: same exon chain; coding sequence starts at an internal ATG, 62 codons after the TGA.
proteins human P62341; C. intestinalis Cint_SELENOT.2.

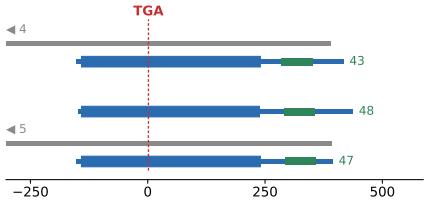
DIO_chr1

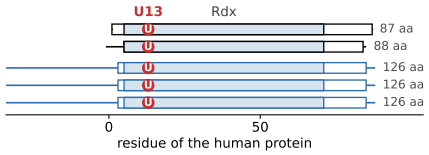




OKI2018 RefSeq XP_081632420.1: coding sequence starts 4 codons after the TGA; model corrected for the assembly (low-quality protein). Gold standard: mapped from the transcript (the assembly lacks 7 bases of the coding sequence and has 6 others).
OSKA2016 ANISEED S37.g16254.01.t: TGA exon skipped; joined to 5 exons upstream of the gene.
OdB3 GenBank a CBY06720.1: ends at the TGA (read as stop); b CBY06719.1: starts 4 codons after the TGA.
proteins human P55073; C. intestinalis Cint_DIO.2.

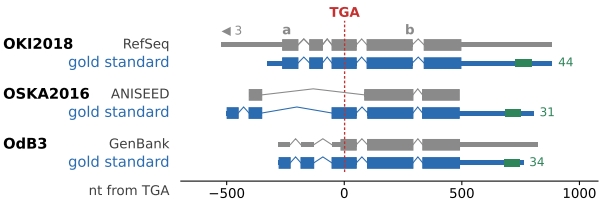
SELENOW_PAR

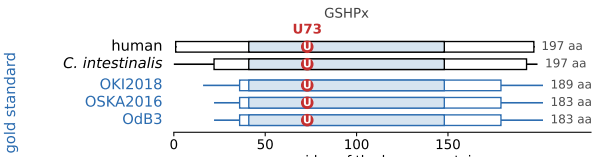




OKI2018 RefSeq XP_081636423.1: mRNA of another protein (592 aa), with the gene in its 3′ UTR.
OSKA2016 no ANISEED model. Gold standard: mapped from the transcript (the assembly lacks 2 bases of the coding sequence).
OdB3 GenBank CBY23864.1: mRNA of another protein (589 aa), with the gene in its 3′ UTR.
proteins human P63302; C. intestinalis Cint_SELENOW.1.

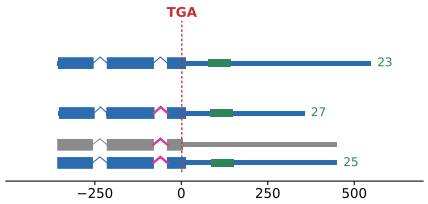
GPX_PAR

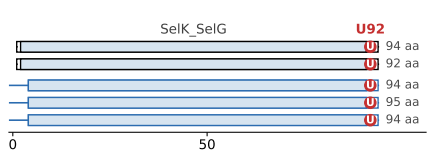




OKI2018 RefSeq a XR_014134294.1: non-coding RNA that starts upstream and ends in the TGA exon; b XP_081635498.1: selenocysteine declared; same model as the gold standard.
OSKA2016 ANISEED S25.g14813.01.t: TGA exon skipped; starts in the second exon.
OdB3 GenBank CBY25160.1: starts 14 nt before the TGA, in another frame for 41 codons. Gold standard: mapped from the transcript (one extra base in the assembly).
proteins human P36969; C. intestinalis Cint_GPX.5.

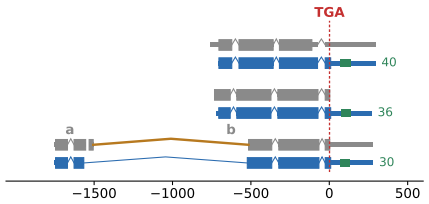
SELENOK

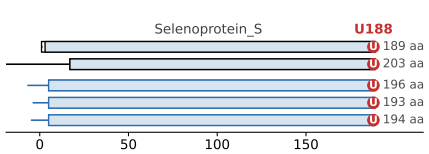




OKI2018 no RefSeq model.
OSKA2016 no ANISEED model.
OdB3 GenBank CBY09056.1: same exon chain; TGA read as stop.
proteins human Q9Y6D0; C. intestinalis Cint_SELENOK.2.

SELENOS





OKI2018 RefSeq XP_081639144.1: same exon chain; frame lost where the assembly lacks 13 bases; ends before the TGA. Gold standard: mapped from the transcript (the assembly lacks 13 bases of the coding sequence).
OSKA2016 ANISEED S13.g10968.01.t: starts 9 codons upstream; TGA read as stop.
OdB3 GenBank a CBY08859.1: N-terminal part; ends in an intron; b CBY08858.1: other first exon; TGA read as stop.
proteins human Q9BQE4; C. intestinalis Cint_SELENOS.1.

exon: coding (thick), UTR (thin) — public model — gold-standard model — coding sequence in another frame
selenocysteine TGA (x axis: nucleotides from it, gene orientation) — 37 SECIS element and bit score (current model)
◂n number of exons beyond the drawn region — a b public models of the same row, as named in the paragraph
intron splice sites (drawn length unchanged): GT-AG (colour of the model) — GA-AG — GC-AG — GG-AG — other site
Pfam domain of the first reference — U selenocysteine aligned to one of the first reference — © other residue aligned to it
residues not aligned to the first reference — human: UniProt; C. intestinalis: stage 1 predictions on its genome (accessions in the paragraphs)